MAT120 Lab_Lab Report - 01

Name: **Student ID:** **Section:**

Instructions:

1. Solve each question using **MATHEMATICA**.
2. Include the **MATHEMATICA Code** and the corresponding output or plot for every question.
3. Provide a brief explanation of the method and results in a Text cell. If you can't do it yet, don't worry and leave it for now.
4. Organize your solutions clearly and ensure that all code and outputs are readable.

Submission: Submit a hard copy of the completed work.

Deadline: October 17,2026

A. CONTINUITY

Let

$$f(x) = \begin{cases} x + 2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

Is f continuous at x = 0? Justify your answer using the left-hand limit, the right-hand limit and f (0).

B. TRANSFORMATIONS

Plot the following functions on the same graph:

$$y = |x|,\ y = |x - 3|,\ y = |x| + 2.$$

Describe how each graph is obtained from $y = |x|$.

C. RATIONAL FUNCTION

Let

$$f(x) = \frac{x^2 - 3}{x - 1}$$

Find the domain and the intercepts of f(x), then plot the function.

D. CREATE

Sketch or plot a function that is continuous everywhere except at x = 2. State whether your discontinuity is removable, jump or infinite.

E. EXPLAIN

In your own words, explain the difference between a removable discontinuity and a jump discontinuity.